

Foundation (Jalapeno)

- Q1.** What is the graph of $y > 2x - 3$?
(A) A solid line with shaded region above
(B) A dashed line with shaded region above
(C) A solid line with shaded region below
(D) A dashed line with shaded region below
(E) No shaded region
- Q2.** The inequality $x + y \leq 4$ has a:
(A) Solid boundary line
(B) Dashed boundary line
(C) No boundary line
(D) Shaded region only
(E) None of the above
- Q3.** Which of the following inequalities has a solution region that is unbounded?
(A) $x > 2$ and $y < 3$
(B) $x < 2$ and $y > 3$
(C) $x \geq 2$ and $y \leq 3$
(D) $x \leq 2$ and $y \geq 3$
(E) $x = 2$ and $y = 3$
- Q4.** What is the equation of the boundary line for $y > x - 2$?
(A) $y = x - 2$
(B) $y = x + 2$
(C) $y = -x - 2$
(D) $y = -x + 2$
(E) $x = y - 2$
- Q5.** The inequality $y \geq x - 3$ has a:
(A) Solid boundary line
(B) Dashed boundary line
(C) No boundary line
(D) Shaded region only
(E) None of the above
- Q6.** The solution region for $x + 2y \leq 6$ is:
(A) A triangle with vertices at $(0, 0)$, $(6, 0)$, and $(0, 3)$
(B) A triangle with vertices at $(0, 0)$, $(0, 3)$, and $(3, 0)$
(C) A rectangle with vertices at $(0, 0)$, $(6, 0)$, $(6, 3)$, and $(0, 3)$
(D) A trapezoid with vertices at $(0, 0)$, $(6, 0)$, $(0, 3)$, and $(6, 3)$
(E) None of the above
- Q7.** Which inequality represents the region below the line $y = -2x + 1$?
(A) $y \leq -2x + 1$
(B) $y \geq -2x + 1$
(C) $y > -2x + 1$
(D) $y < -2x + 1$
(E) $x > -2y + 1$
- Q8.** The inequality $y < x^2$ has a:
(A) Solid boundary line
(B) Dashed boundary line
(C) No boundary line
(D) Shaded region only
(E) None of the above

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $y > x - 3$ and identify the solution region.
- Q10.** Find the equation of the boundary line for $x + 2y \geq 4$ and determine if it is solid or dashed.

Chef's Tips

- Emphasize the difference between solid and dashed boundary lines.
- Use graph paper to help students visualize the solution regions.
- Encourage students to check their work by plugging in test points.